HDI Predictor (Using Machine Learning)

Project Description:

HDI Predictor – Human Development Index Prediction System is an AI-powered web application that predicts the Human Development Index (HDI) category of a country using
Machine Learning. The system analyzes four key socio-economic indicators: **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita** to classify countries into **Low, Medium, High, or Very High** HDI
categories. The application uses the **Random Forest Machine Learning algorithm** to generate accurate predictions and is deployed using the **Flask** framework. It also provides real-time prediction results, confidence scores, interactive data visualizations, AI-generated development insights, user authentication, and prediction history through a user-friendly web interface.

Problem Statement

Traditional HDI analysis relies on statistical reports and manual interpretation of multiple socio-economic indicators, making the process time-consuming and difficult to analyze. There is a need for an intelligent, data-driven system that can accurately predict a country's HDI category based on key development indicators. This project automates the prediction process using Machine Learning, enabling faster analysis, better visualization, and improved decision-making.

Scenarios:

Scenario 1: Predicting High Human Development

A user enters a country's Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita into the application. The trained Random Forest model analyzes these indicators and predicts the country's HDI category as High, along with confidence score, graphical analysis, and AI-generated recommendations.

Scenario 2: Identifying Development Gaps

A policymaker enters data representing a developing country with moderate education and income levels. The system predicts a Medium HDI category and provides insights highlighting areas such as healthcare, education, or income that require improvement to achieve higher human development.

Scenario 3: Development Analysis for Research

A researcher evaluates multiple countries by entering different socioeconomic indicators. The application predicts the HDI category for each country, stores the prediction history, and displays interactive charts to support comparative analysis and policy planning.

Objectives

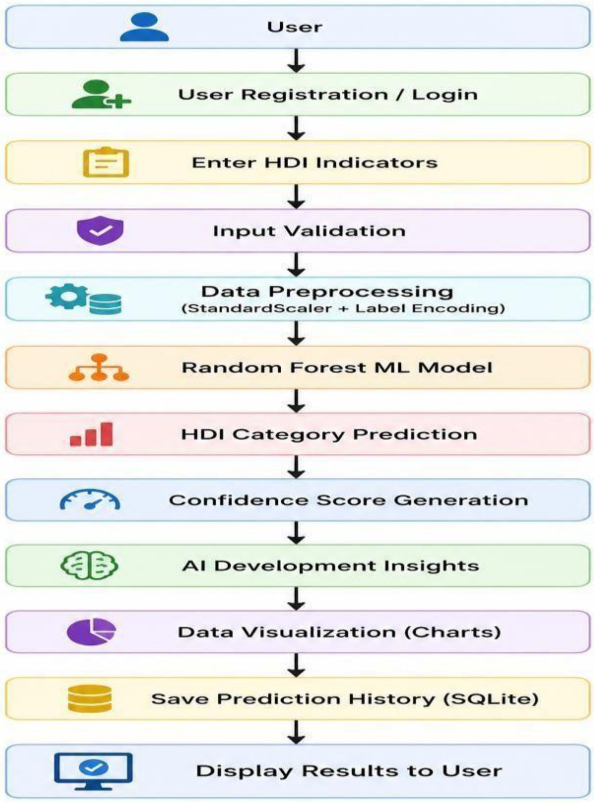
- To develop a Machine Learning model for predicting the Human Development Index (HDI) category.
- To preprocess HDI data and train a **Random Forest** classification model.
- To provide real-time HDI prediction through a Flask-based web application.
- To visualize prediction results using interactive charts and AI-generated insights. To
- implement user authentication and prediction history using SQLite database.

Scope

The scope of this project is limited to predicting the **HDI category** using four socio- economic indicators: **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita**. The application demonstrates an end-to-end Machine Learning workflow, including data preprocessing, model training, prediction, visualization, user authentication, and database management. It is developed for academic and research purposes and is not intended for official government or policy decisionmaking without further validation using real-world datasets.

Technical Architecture:

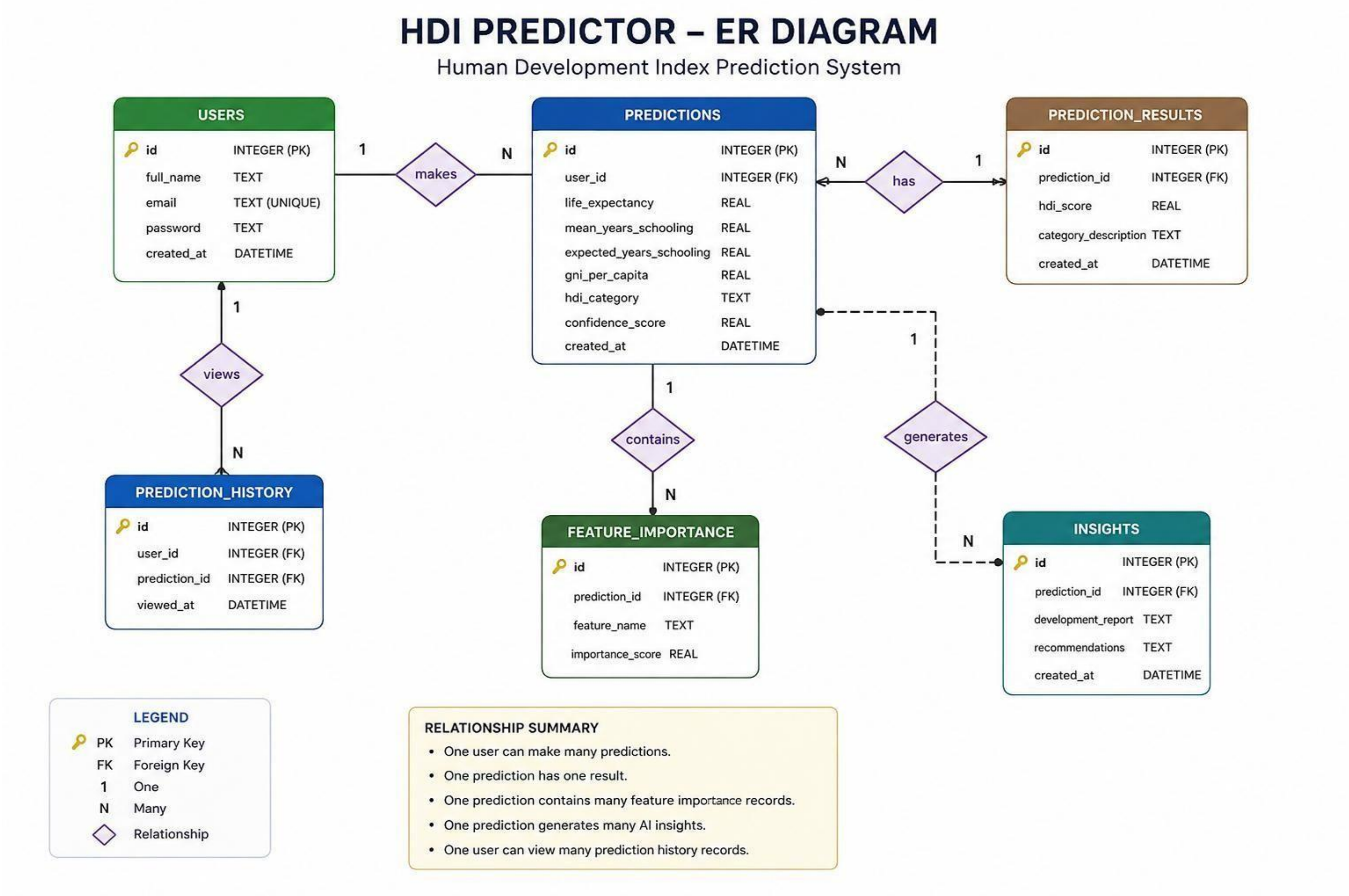
Architecture Flow



ER Diagram

Description:

The **HDI Predictor – Human Development Index Prediction System** follows a Machine Learning-based architecture to provide fast and accurate HDI category predictions. Users securely register and log into the Flask-based web application before entering four socioeconomic indicators: **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita**. The input data is validated and preprocessed before being passed to the trained **Random Forest Machine Learning model**. The model predicts the country's HDI category (**Low, Medium, High, or Very High**) and generates a confidence score. The prediction results are displayed along with interactive charts, AI-generated development insights, and recommendations. Each prediction is stored in the SQLite database, allowing users to access their prediction history through a secure dashboard.



Pre-requisites

- Python Programming (Python 3.x)
- Machine Learning Fundamentals (Scikit-learn)
- Flask Web Framework
- HTML5, CSS3, JavaScript
- Bootstrap 5
- SQLite Database
- Pandas & NumPy
- Chart.js (Data Visualization)
- Git & GitHub
- Visual Studio Code

Project Workflow:

Milestone 1: Data Collection & Understanding

- Activity 1.1: Import the hdi_dataset.csv dataset.
- Activity 1.2: Analyze dataset structure, columns, and data types.
- Activity 1.3: Perform Exploratory Data Analysis (EDA) to understand feature distribution.
- Activity 1.4: Identify input and target variables.

Milestone 2: Data Preprocessing

- Activity 2.1: Check for missing values.
- Activity 2.2: Remove duplicate records.
- Activity 2.3: Encode HDI category labels using Label Encoder.
- Activity 2.4: Select the required input features.
- Activity 2.5: Split the dataset into Training and Testing datasets.
- Activity 2.6: Apply preprocessing where required.

Milestone 3: Model Building

- Activity 3.1: Train the Random Forest Classifier.
- Activity 3.2: Optimize model parameters.
- Activity 3.3: Save the trained model using Joblib.

Milestone 4: Model Evaluation

- Activity 4.1: Evaluate the model using Accuracy.
- Activity 4.2: Generate Confusion Matrix.
- Activity 4.3: Calculate Precision, Recall, and F1-Score.
- Activity 4.4: Select the Random Forest model based on the best performance.

Milestone 5: Deployment

- Activity 5.1: Save the trained model (model.pkl).
- Activity 5.2: Develop the Flask web application.
- Activity 5.3: Create responsive HTML, CSS, and JavaScript frontend.
- Activity 5.4: Integrate the trained model with the Flask backend.
- Activity 5.5: Implement Login, Registration, Prediction History, and Dashboard.
- Activity 5.6: Test the application and verify end-to-end prediction functionality.

Methodology

Dataset Description

Attribute	**Details**
Dataset Name	hdi_dataset.csv
Type	Structured dataset
Target Variable	HDI Category (Low, Medium, High, Very High)
Feature Types	Numerical Features

Data Preprocessing Steps

- Load the HDI dataset using Pandas.
- Inspect dataset using **head(), info(), describe()**.
- Check for missing values.
- Remove duplicate records.
- Encode HDI category labels using Label Encoder. Select
- input features:
  - Life Expectancy
  - Mean Years of Schooling
  - Expected Years of Schooling
  - GNI per Capita
- Split the dataset into Training (80%) and Testing (20%).
- Prepare data for Machine Learning model training.

Algorithms Used

- Random Forest Classifier

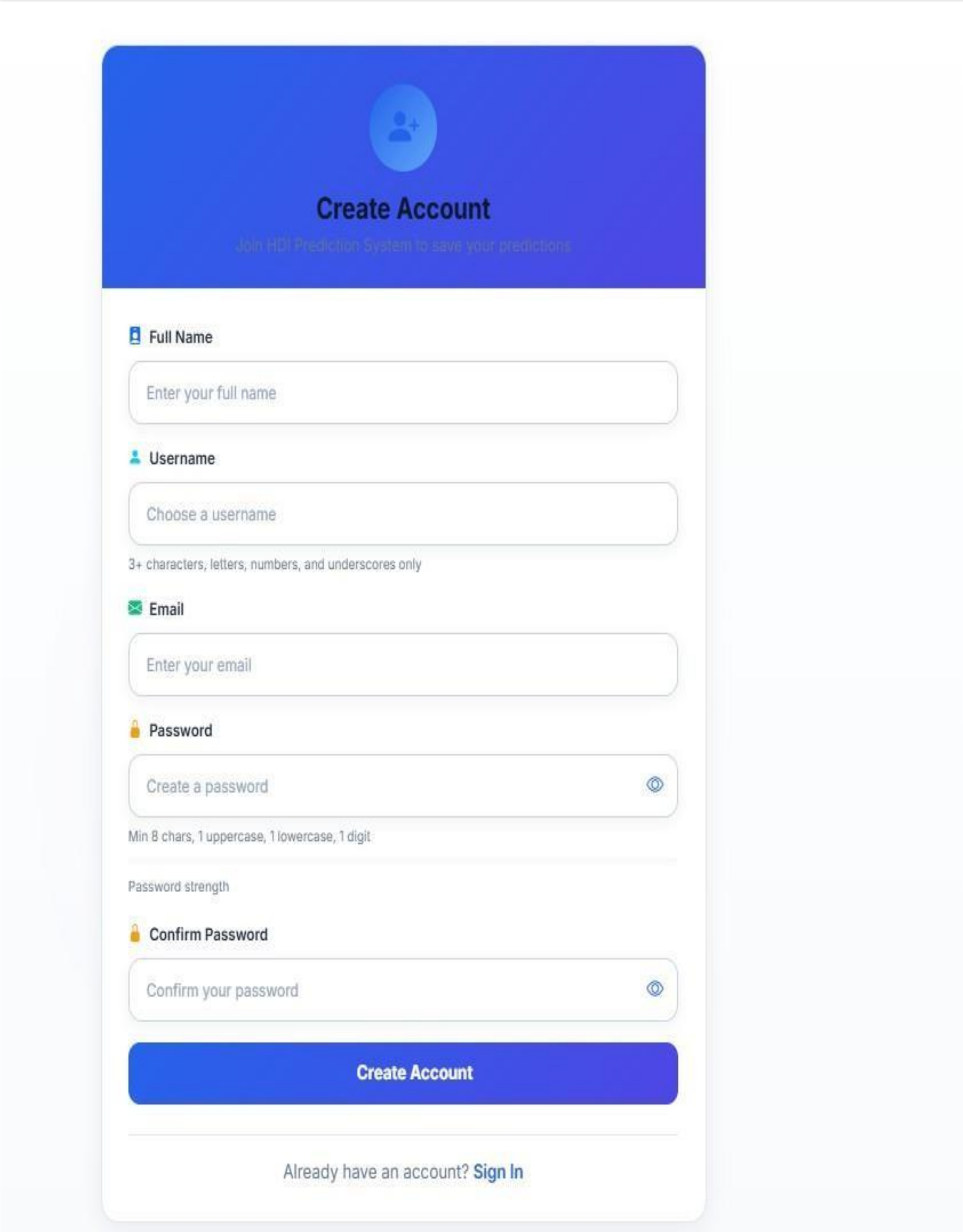
Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

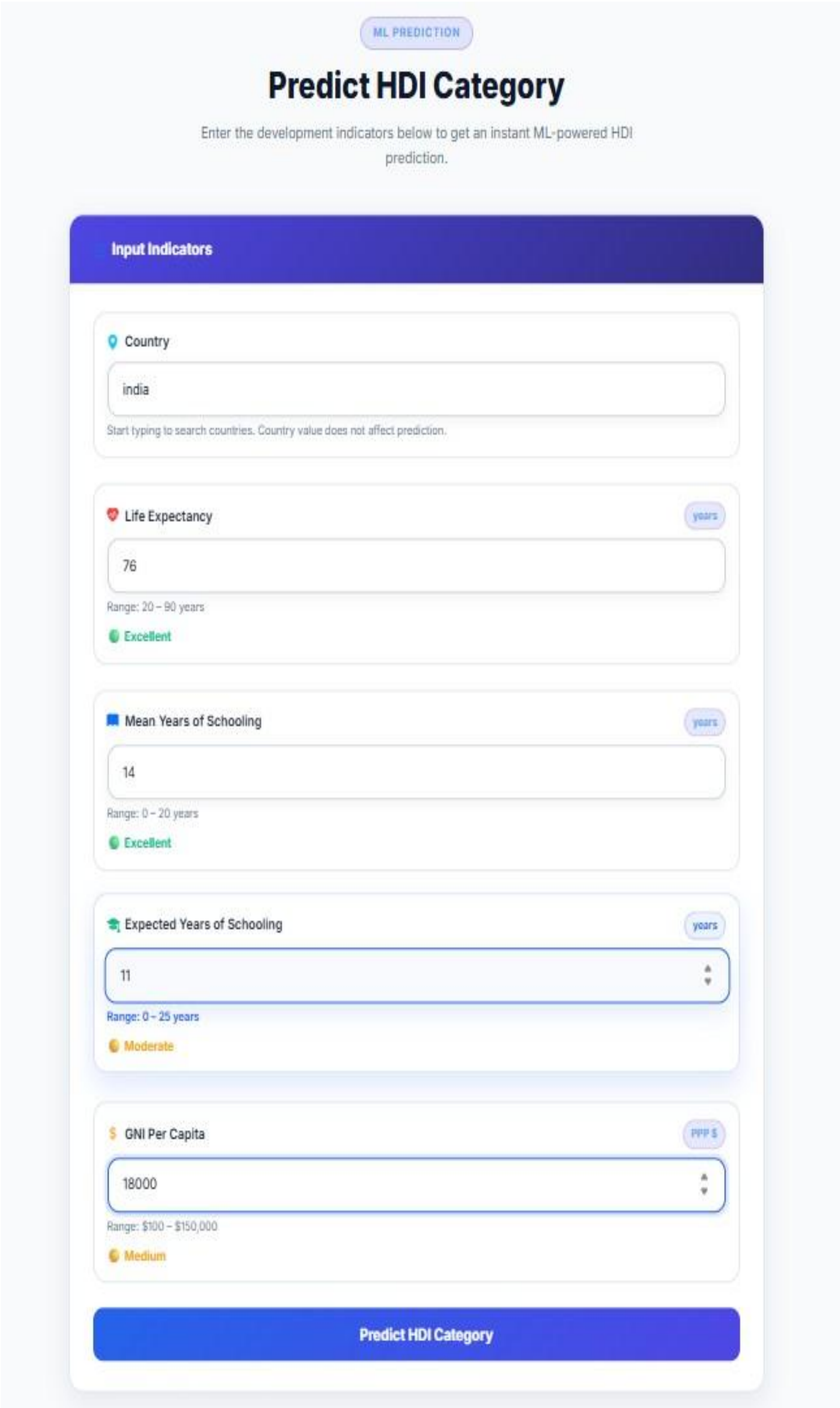
Web Application

The trained Random Forest Machine Learning model has been deployed as an interactive web application called HDI Predictor – Human Development Index Prediction System. The application is developed using Flask (Backend) and HTML, CSS, JavaScript, Bootstrap, and Chart.js
(Frontend).
Users can securely register and log in, enter the four HDI indicators, and instantly receive the predicted HDI category along with a confidence score,
interactive visualizations, AI-generated development insights, and personalized recommendations. Each prediction is automatically stored in the Prediction History page using an SQLite database, allowing users to review previous predictions at any time.

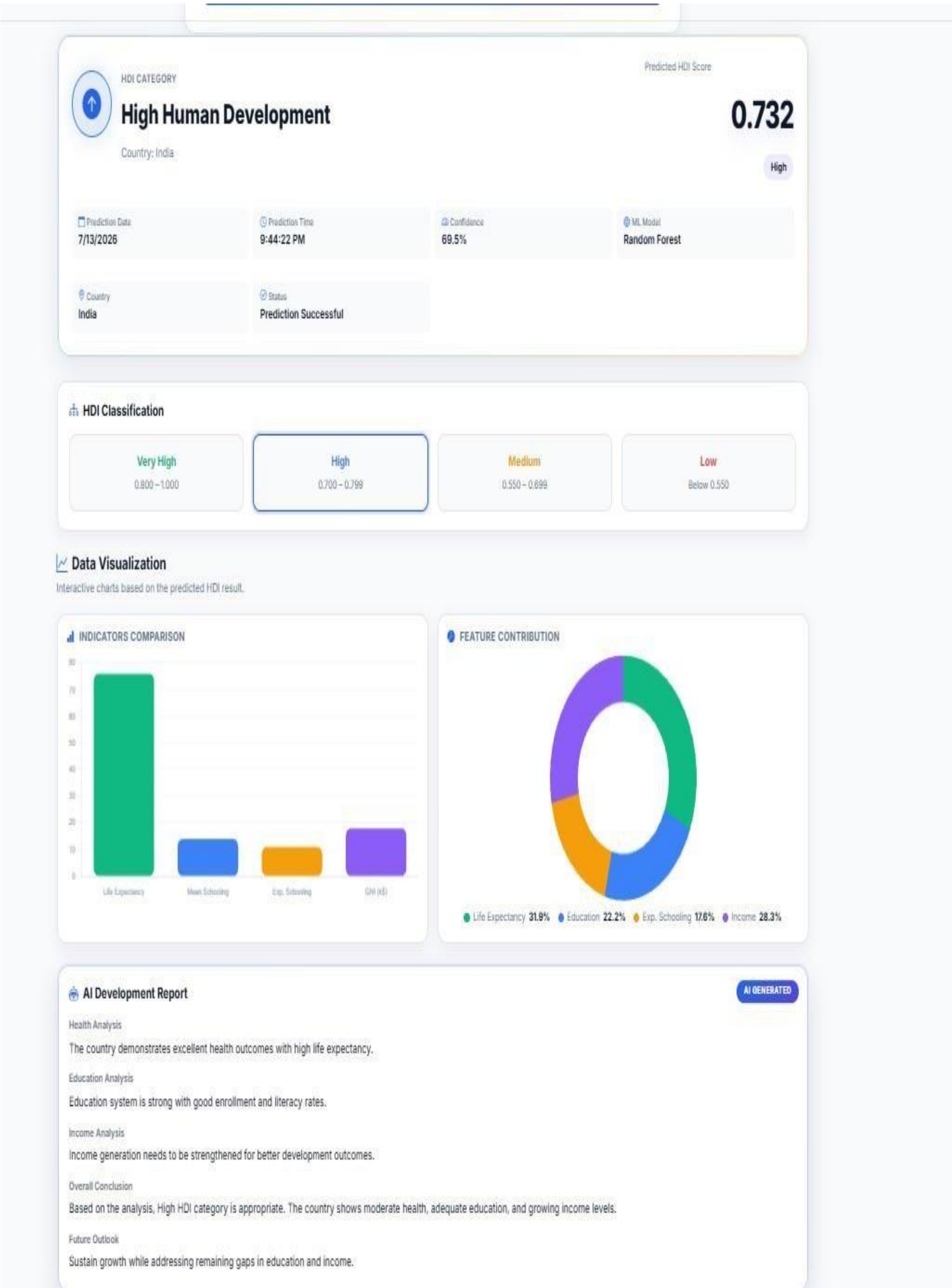
Register Form



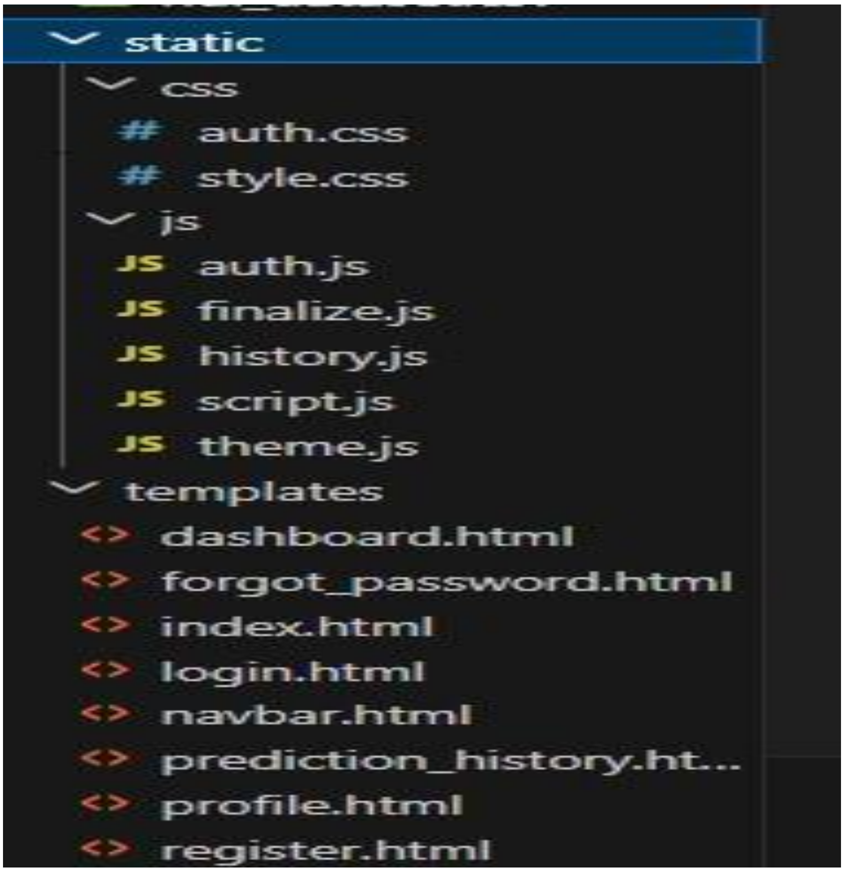
Prediction form:



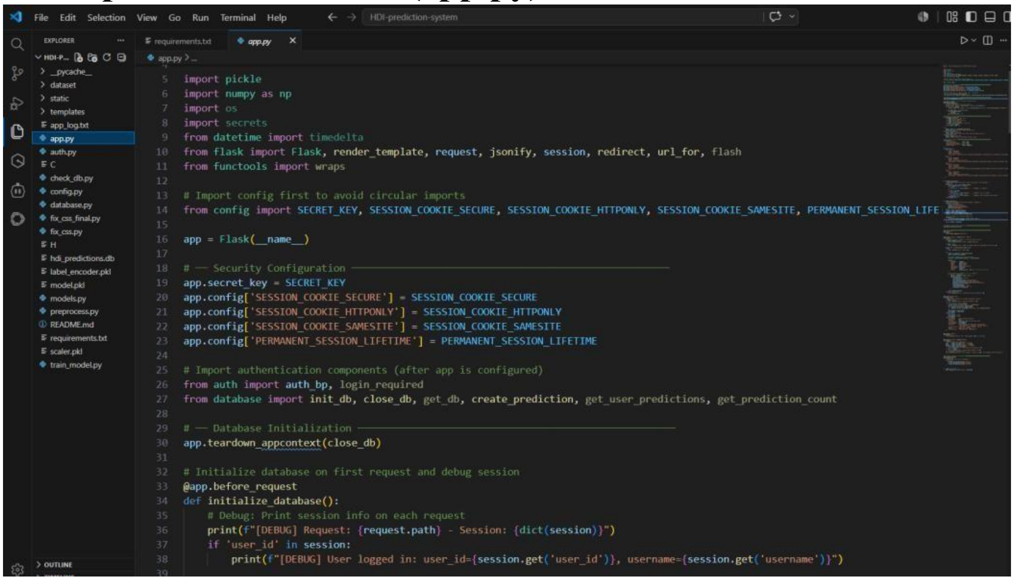
Prediction Result Form:



Frontend Implementation



Backend Implementation Flask(app.py)



Conclusion and Future Scope

Conclusion

- The HDI Predictor – Human Development Index Prediction System successfully demonstrates the application of Machine Learning in analysing and predicting a country's development level based on key socio-economic indicators such as Life Expectancy, Education, and GNI per Capita. The Random Forest model effectively classifies countries into Low, Medium, High, and Very High HDI categories and provides real-time predictions through a Flask-based web application.
- The system integrates Machine Learning, Full-Stack Web Development, Database Management, and Data Visualization to deliver an interactive and user-friendly platform. Features such as user authentication, prediction history, confidence scores, charts, and AI-based development insights improve usability and help users understand factors influencing human development.
- Overall, this project demonstrates how data-driven approaches can support development analysis, decision-making, and better understanding of socio-economic progress.
- **Future Scope**
- The HDI Prediction System can be enhanced with the following improvements:
- • Integrating larger and more updated global development datasets to improve prediction accuracy.
- Deploying the application on cloud platforms such as AWS, Azure, or Google Cloud for scalability.
- Implementing advanced Machine Learning and Deep Learning models for improved predictions.
- Adding Explainable AI (XAI) techniques such as SHAP and LIME to explain model decisions.
- Integrating real-time data sources from international development databases.
- Developing a mobile application for easier access to HDI analysis.
- Implementing advanced analytics and comparison features between countries.
- Continuously retraining the model with new data to improve performance over time.
- Adding role-based access control for administrators and researchers.
- Extending the system for applications such as economic forecasting and policy recommendation systems.